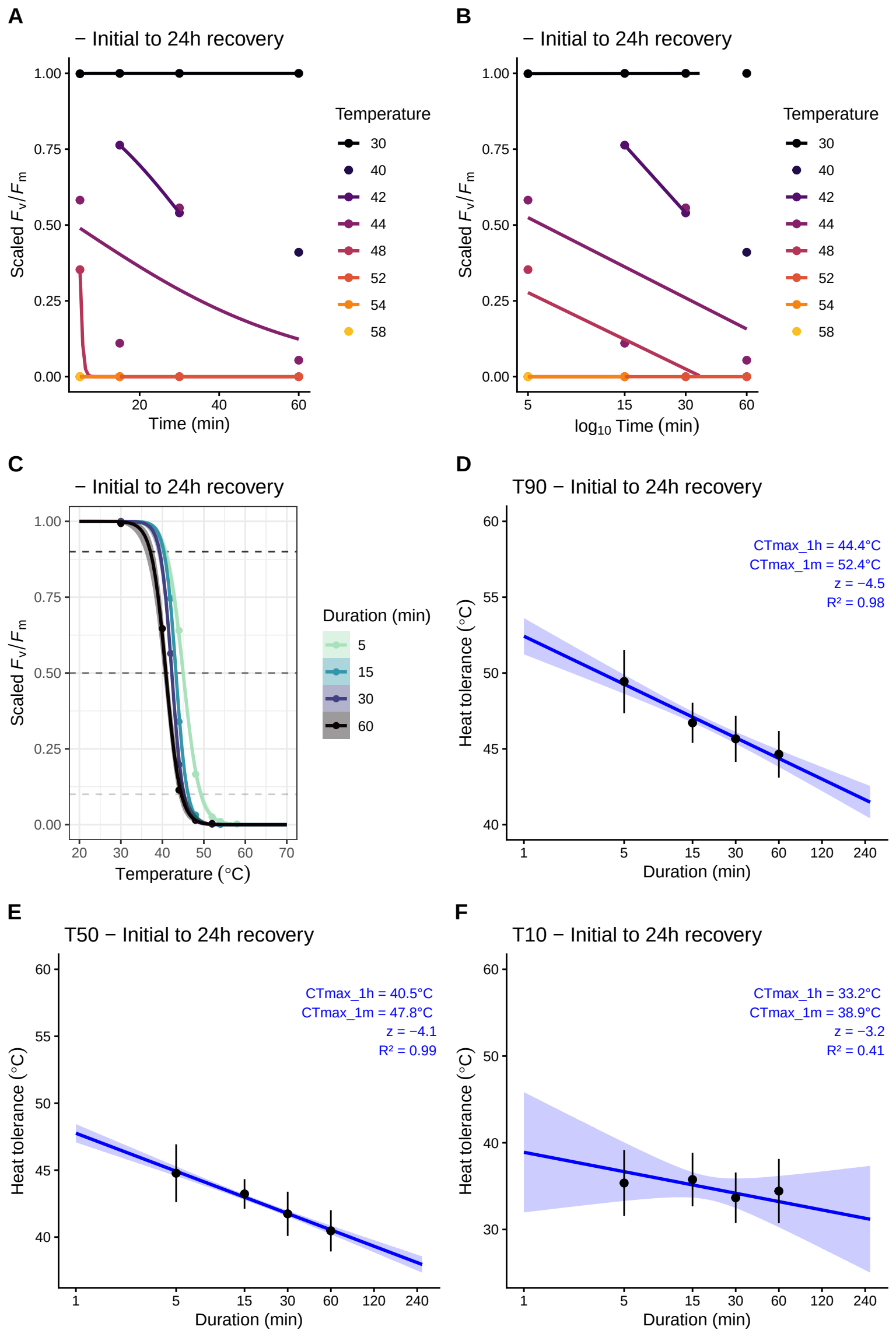
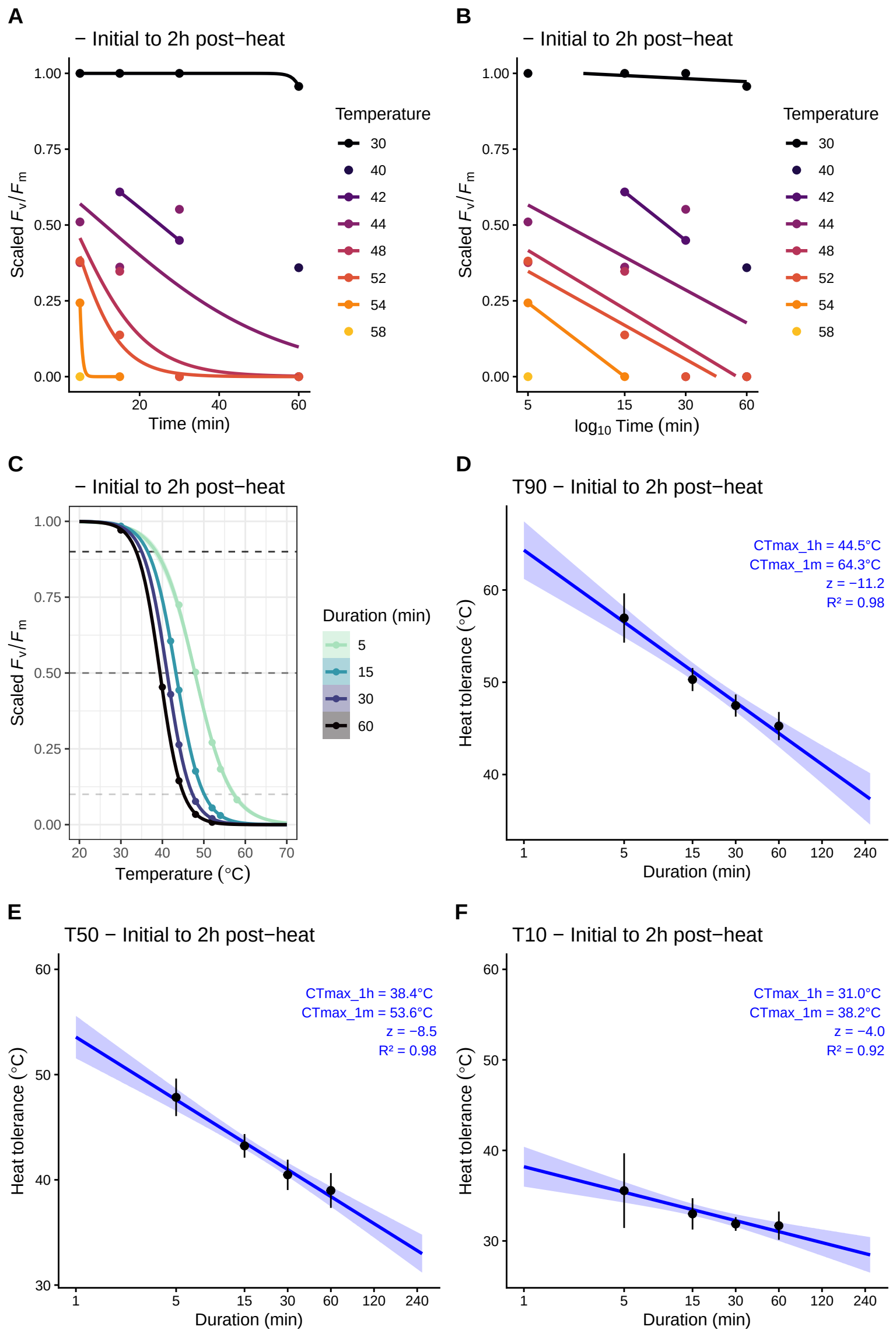
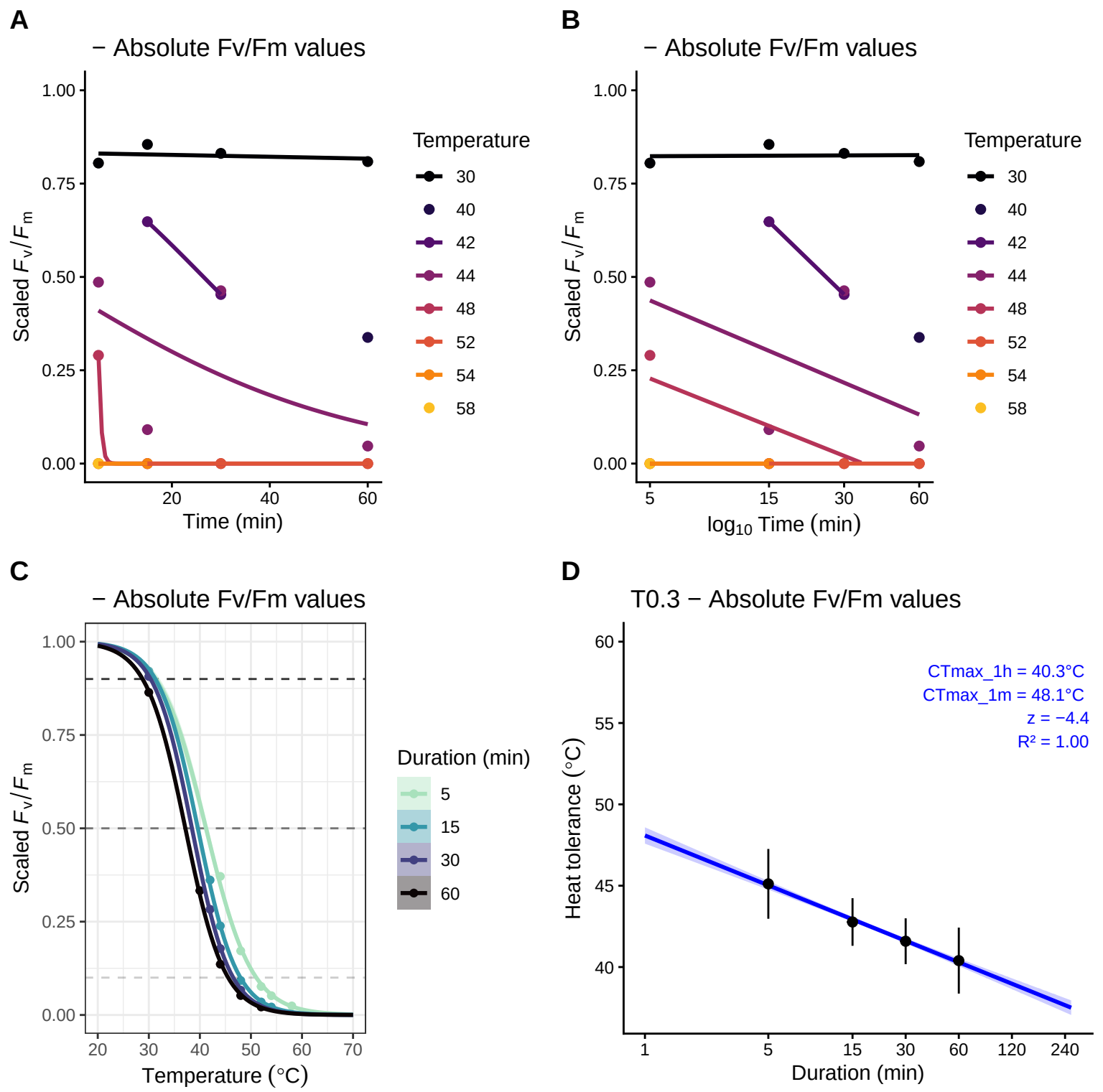
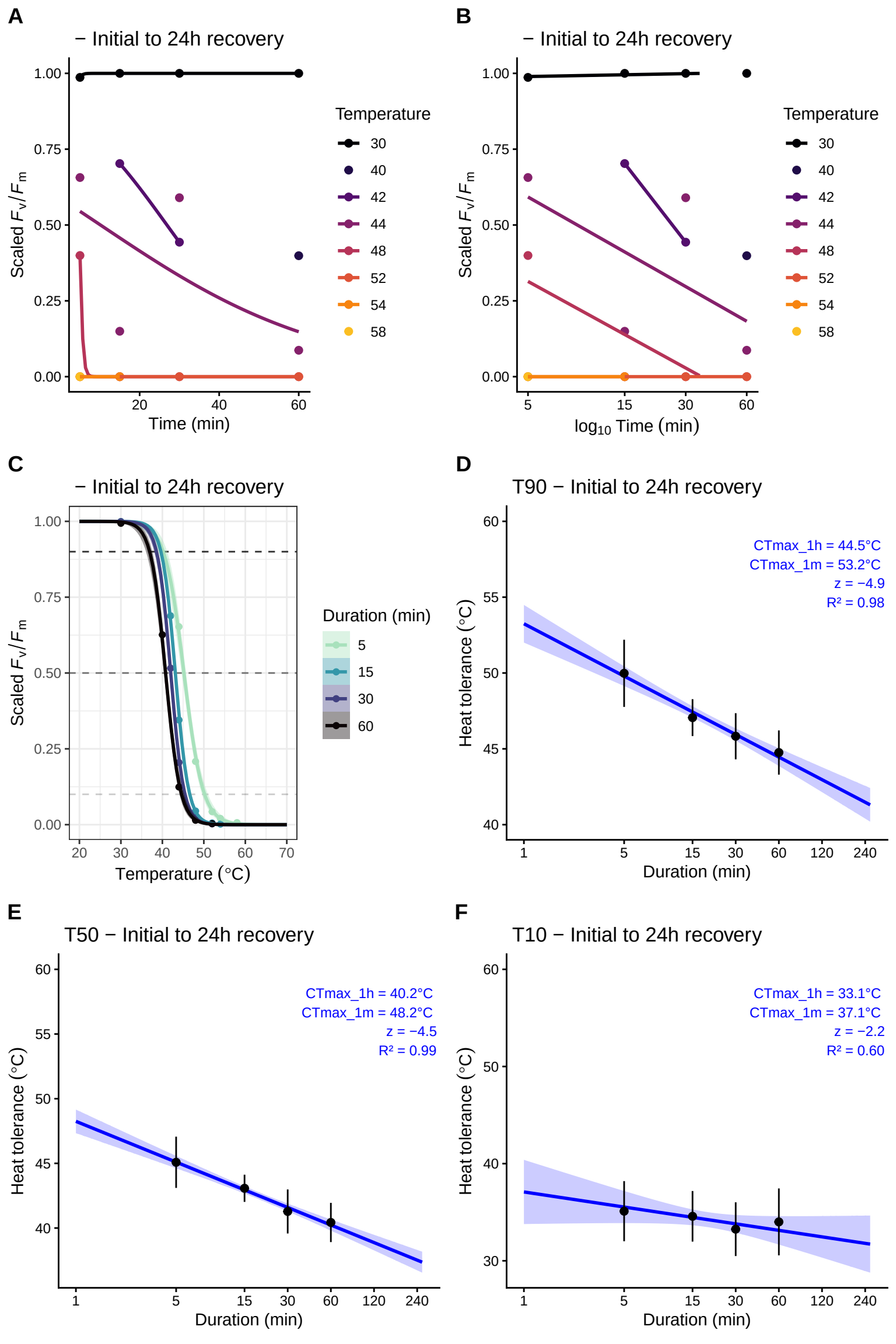
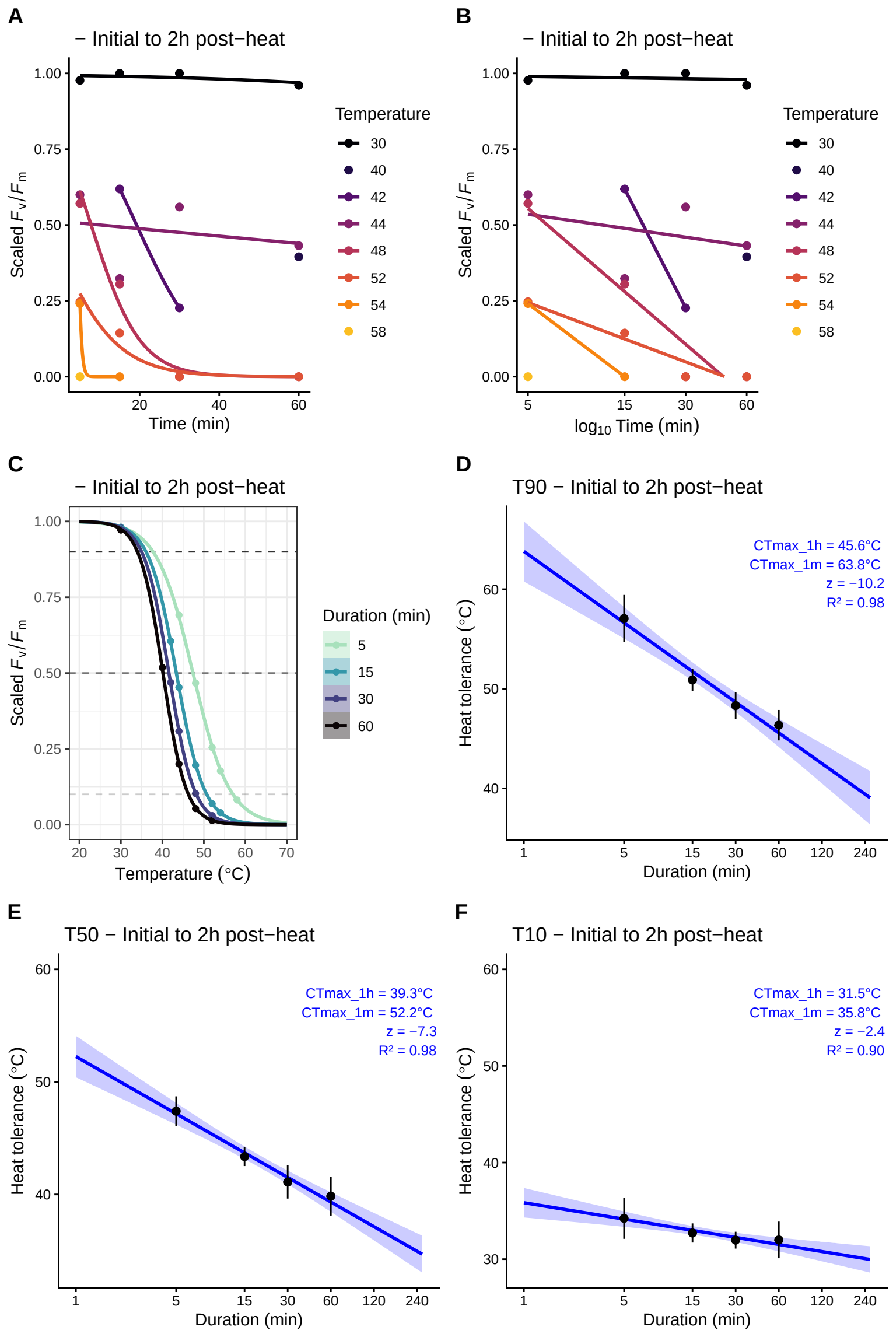
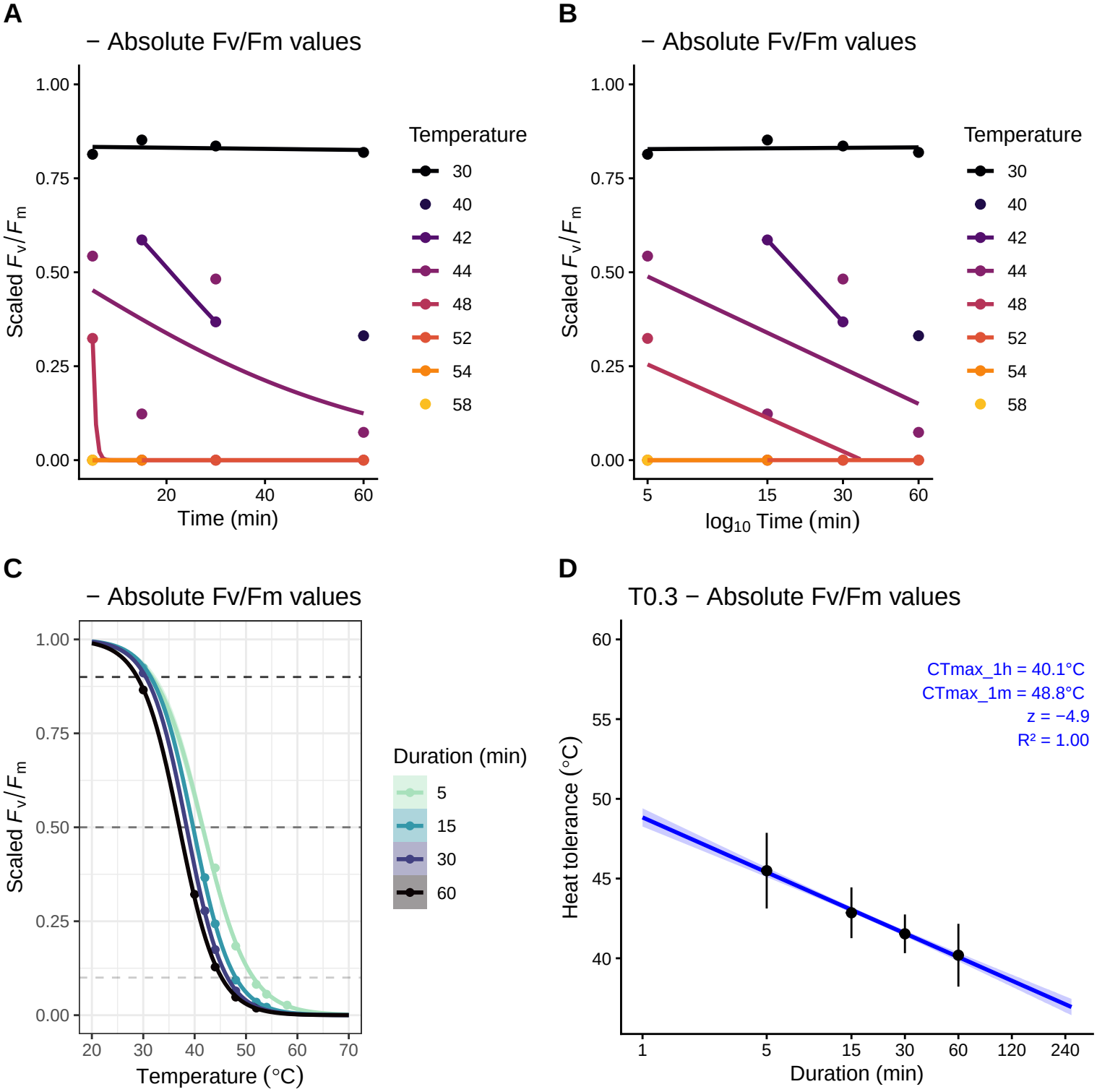

A
− Initial to 24h recovery
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Initial to 24h recovery
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Initial to 24h recovery
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T90 − Initial to 24h recovery
CTmax_1h = 44.4°C
CTmax_1m = 52.4°C
z = −4.5
R² = 0.98
Heat tolerance (°C)
Duration (min)
E
T50 − Initial to 24h recovery
CTmax_1h = 40.5°C
CTmax_1m = 47.8°C
z = −4.1
R² = 0.99
Heat tolerance (°C)
Duration (min)
F
T10 − Initial to 24h recovery
CTmax_1h = 33.2°C
CTmax_1m = 38.9°C
z = −3.2
R² = 0.41
Heat tolerance (°C)
Duration (min)

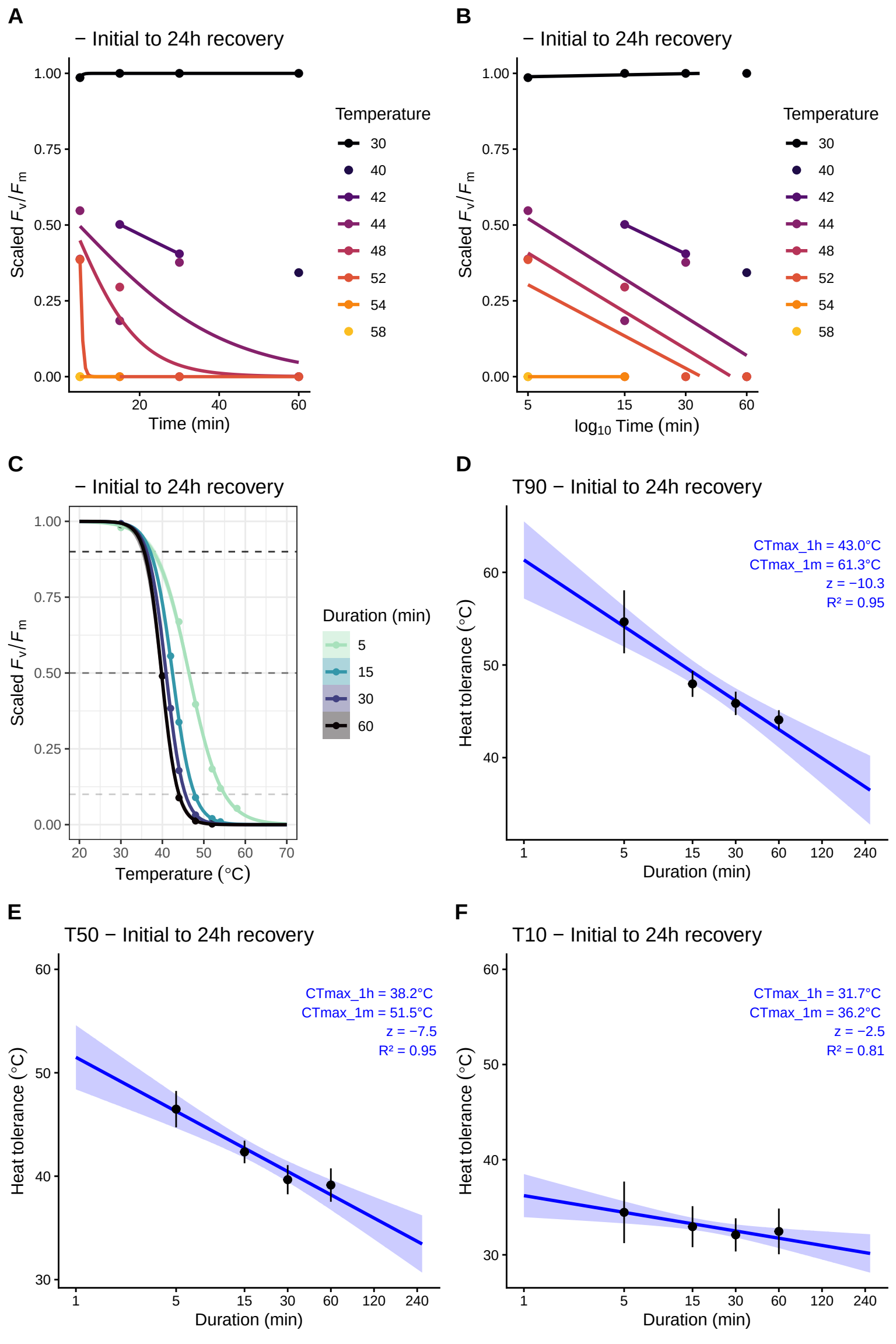

A
− Initial to 2h post−heat
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Initial to 2h post−heat
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Initial to 2h post−heat
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T90 − Initial to 2h post−heat
CTmax_1h = 44.5°C
CTmax_1m = 64.3°C
z = −11.2
R² = 0.98
Heat tolerance (°C)
Duration (min)
E
T50 − Initial to 2h post−heat
CTmax_1h = 38.4°C
CTmax_1m = 53.6°C
z = −8.5
R² = 0.98
Heat tolerance (°C)
Duration (min)
F
T10 − Initial to 2h post−heat
CTmax_1h = 31.0°C
CTmax_1m = 38.2°C
z = −4.0
R² = 0.92
Heat tolerance (°C)
Duration (min)

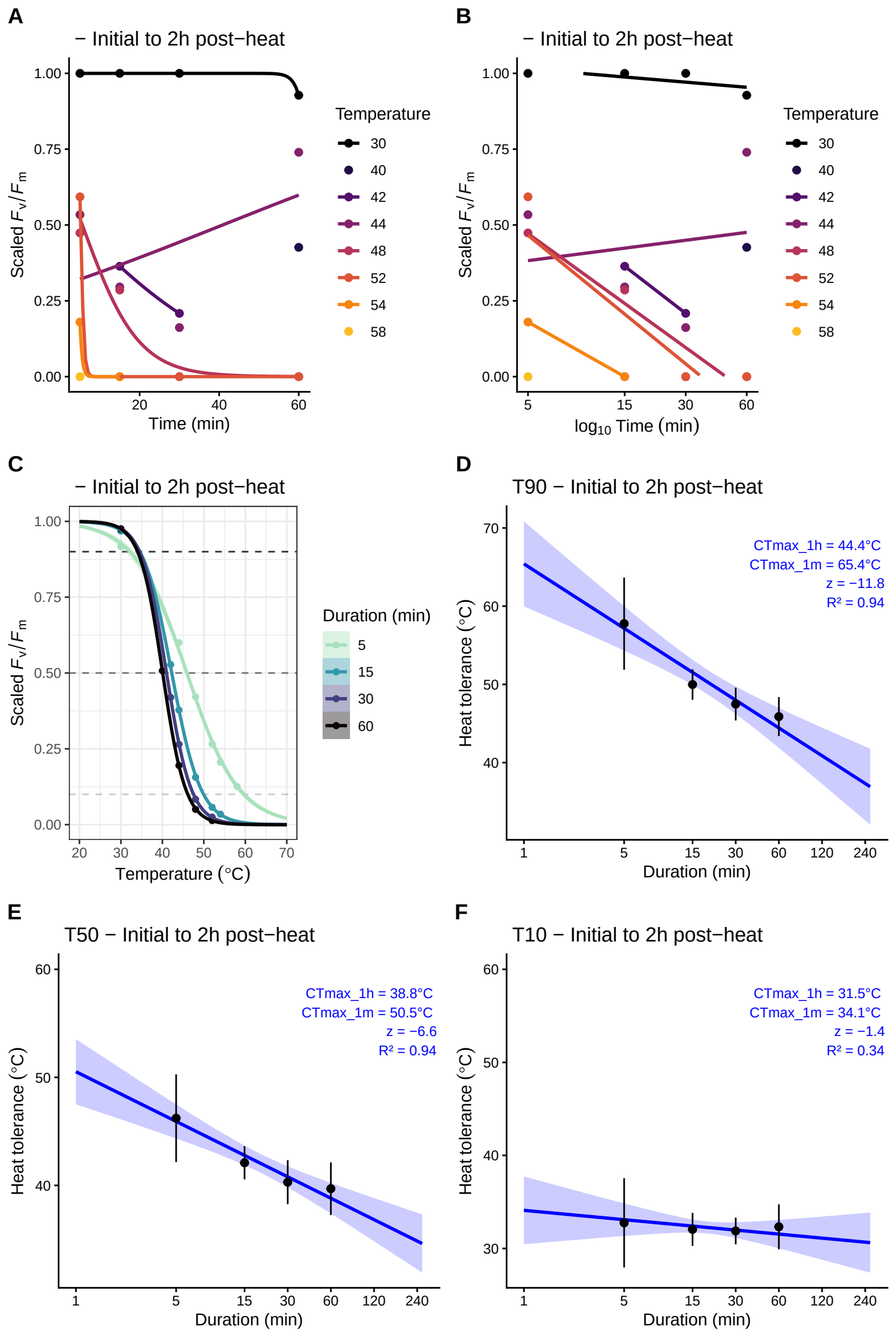

A
− Absolute Fv/Fm values
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Absolute Fv/Fm values
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Absolute Fv/Fm values
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T0.3 − Absolute Fv/Fm values
CTmax_1h = 40.3°C
CTmax_1m = 48.1°C
z = −4.4
R² = 1.00
Heat tolerance (°C)
Duration (min)

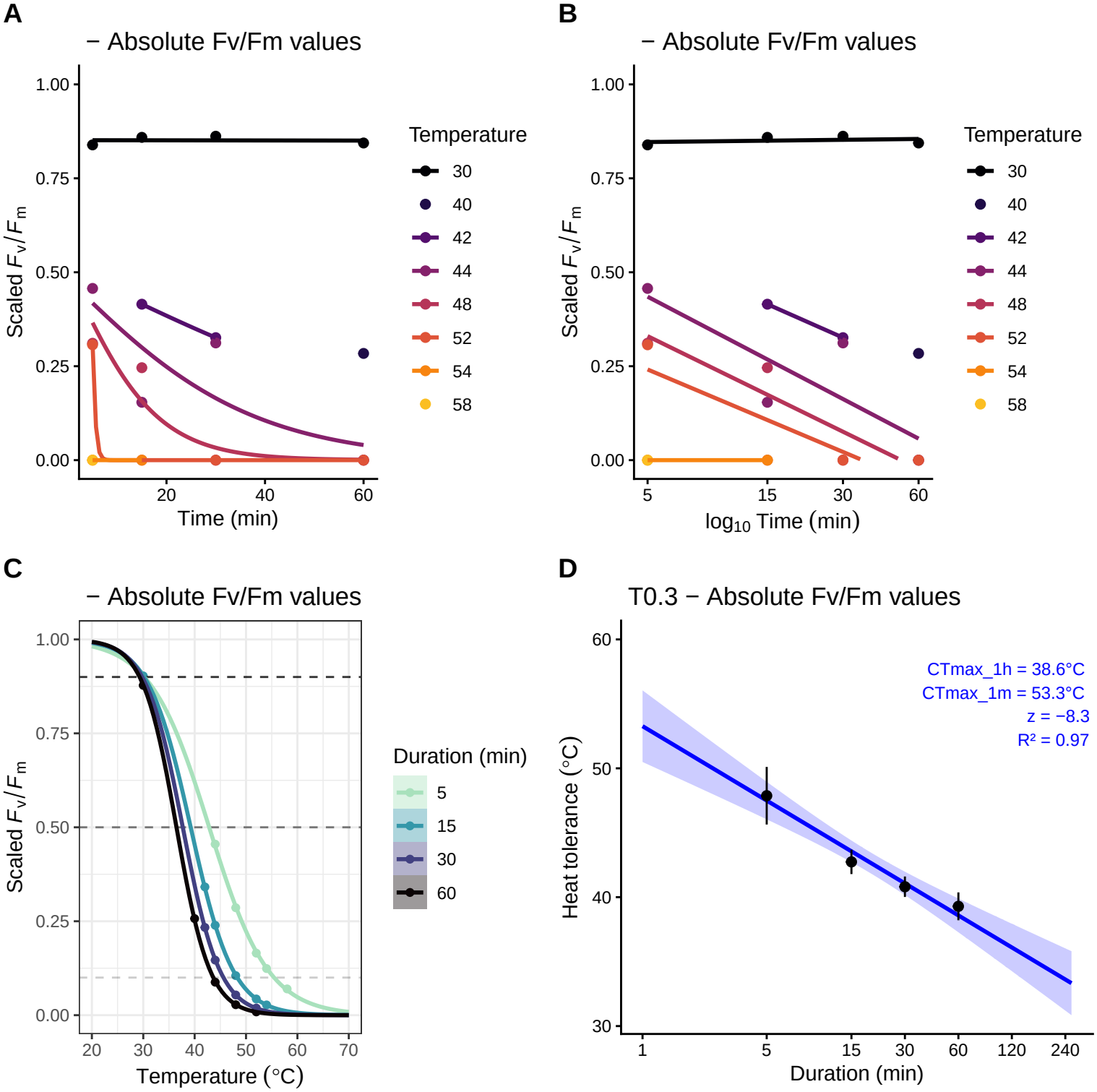

A
− Initial to 24h recovery
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Initial to 24h recovery
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Initial to 24h recovery
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T90 − Initial to 24h recovery
CTmax_1h = 44.5°C
CTmax_1m = 53.2°C
z = −4.9
R² = 0.98
Heat tolerance (°C)
Duration (min)
E
T50 − Initial to 24h recovery
CTmax_1h = 40.2°C
CTmax_1m = 48.2°C
z = −4.5
R² = 0.99
Heat tolerance (°C)
Duration (min)
F
T10 − Initial to 24h recovery
CTmax_1h = 33.1°C
CTmax_1m = 37.1°C
z = −2.2
R² = 0.60
Heat tolerance (°C)
Duration (min)

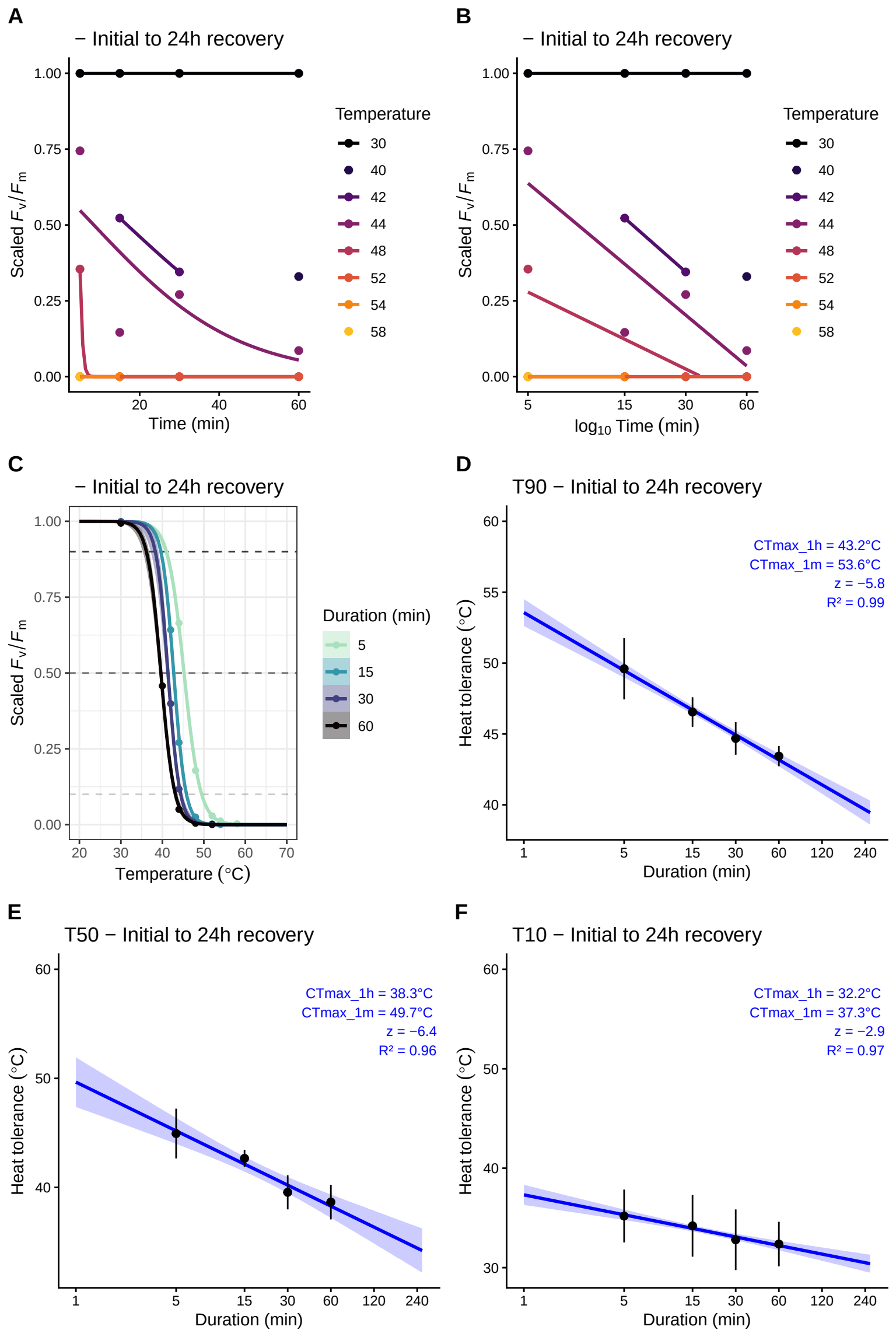

A
− Initial to 2h post−heat
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Initial to 2h post−heat
Scaled F_v/F_m
$\log_{10}$ Time (min)
C
− Initial to 2h post−heat
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T90 − Initial to 2h post−heat
CTmax_1h = 45.6°C
CTmax_1m = 63.8°C
z = −10.2
R² = 0.98
Heat tolerance (°C)
Duration (min)
E
T50 − Initial to 2h post−heat
CTmax_1h = 39.3°C
CTmax_1m = 52.2°C
z = −7.3
R² = 0.98
Heat tolerance (°C)
Duration (min)
F
T10 − Initial to 2h post−heat
CTmax_1h = 31.5°C
CTmax_1m = 35.8°C
z = −2.4
R² = 0.90
Heat tolerance (°C)
Duration (min)

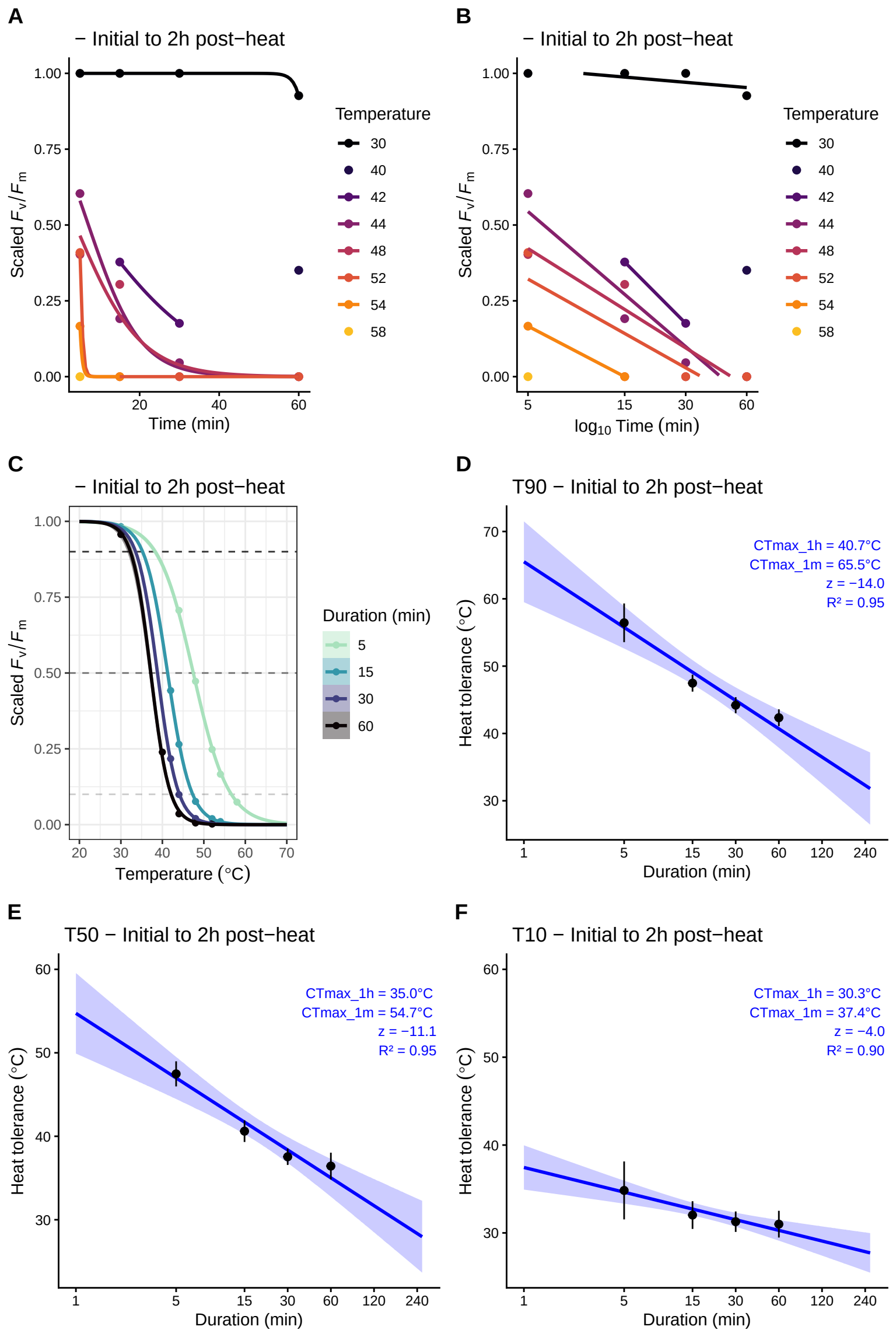

A
− Absolute Fv/Fm values
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Absolute Fv/Fm values
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Absolute Fv/Fm values
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T0.3 − Absolute Fv/Fm values
CTmax_1h = 40.1°C
CTmax_1m = 48.8°C
z = −4.9
R² = 1.00
Heat tolerance (°C)
Duration (min)

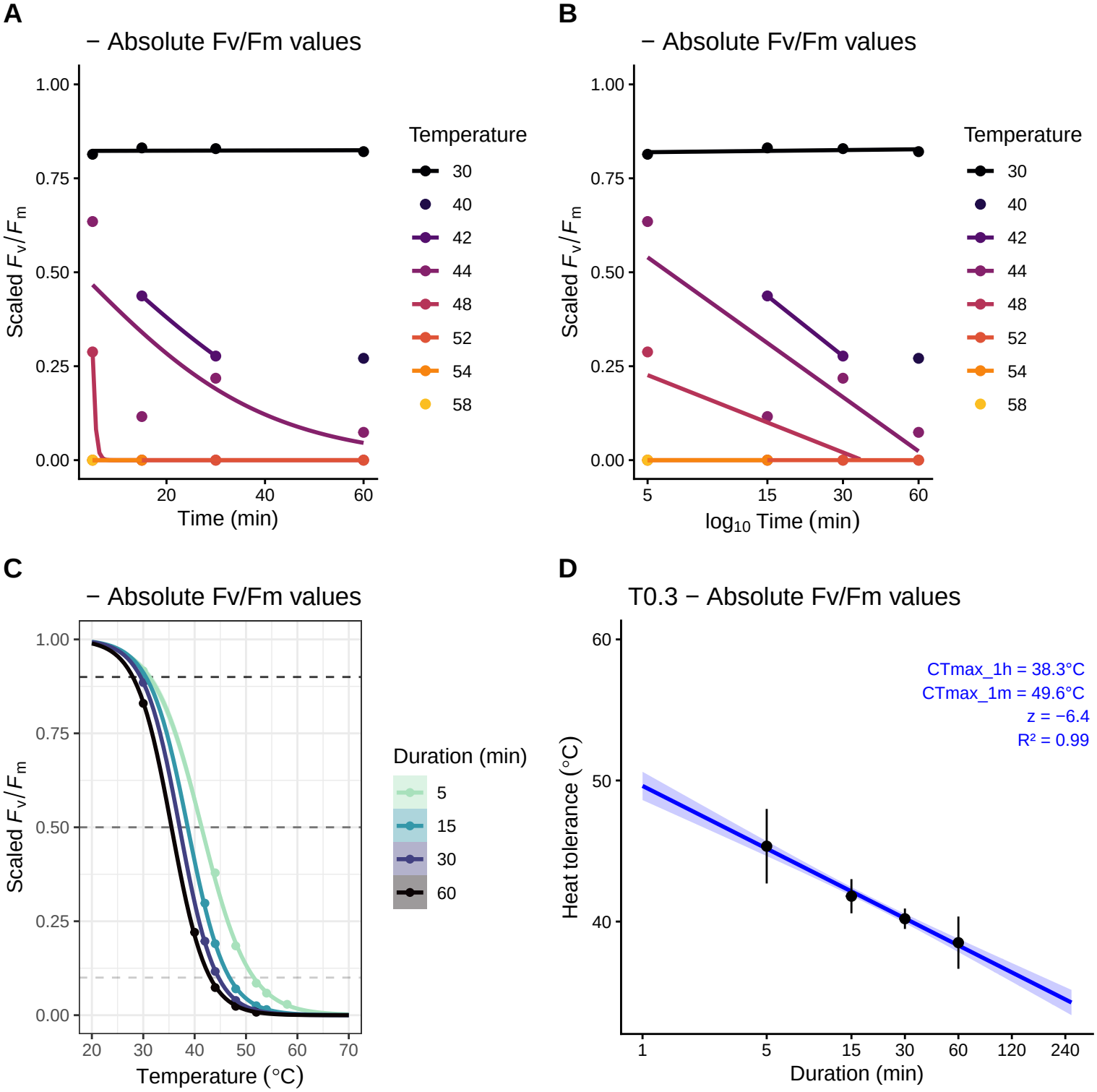

A
− Initial to 24h recovery
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Initial to 24h recovery
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Initial to 24h recovery
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T90 − Initial to 24h recovery
CTmax_1h = 43.0°C
CTmax_1m = 61.3°C
z = −10.3
R² = 0.95
Heat tolerance (°C)
Duration (min)
E
T50 − Initial to 24h recovery
CTmax_1h = 38.2°C
CTmax_1m = 51.5°C
z = −7.5
R² = 0.95
Heat tolerance (°C)
Duration (min)
F
T10 − Initial to 24h recovery
CTmax_1h = 31.7°C
CTmax_1m = 36.2°C
z = −2.5
R² = 0.81
Heat tolerance (°C)
Duration (min)

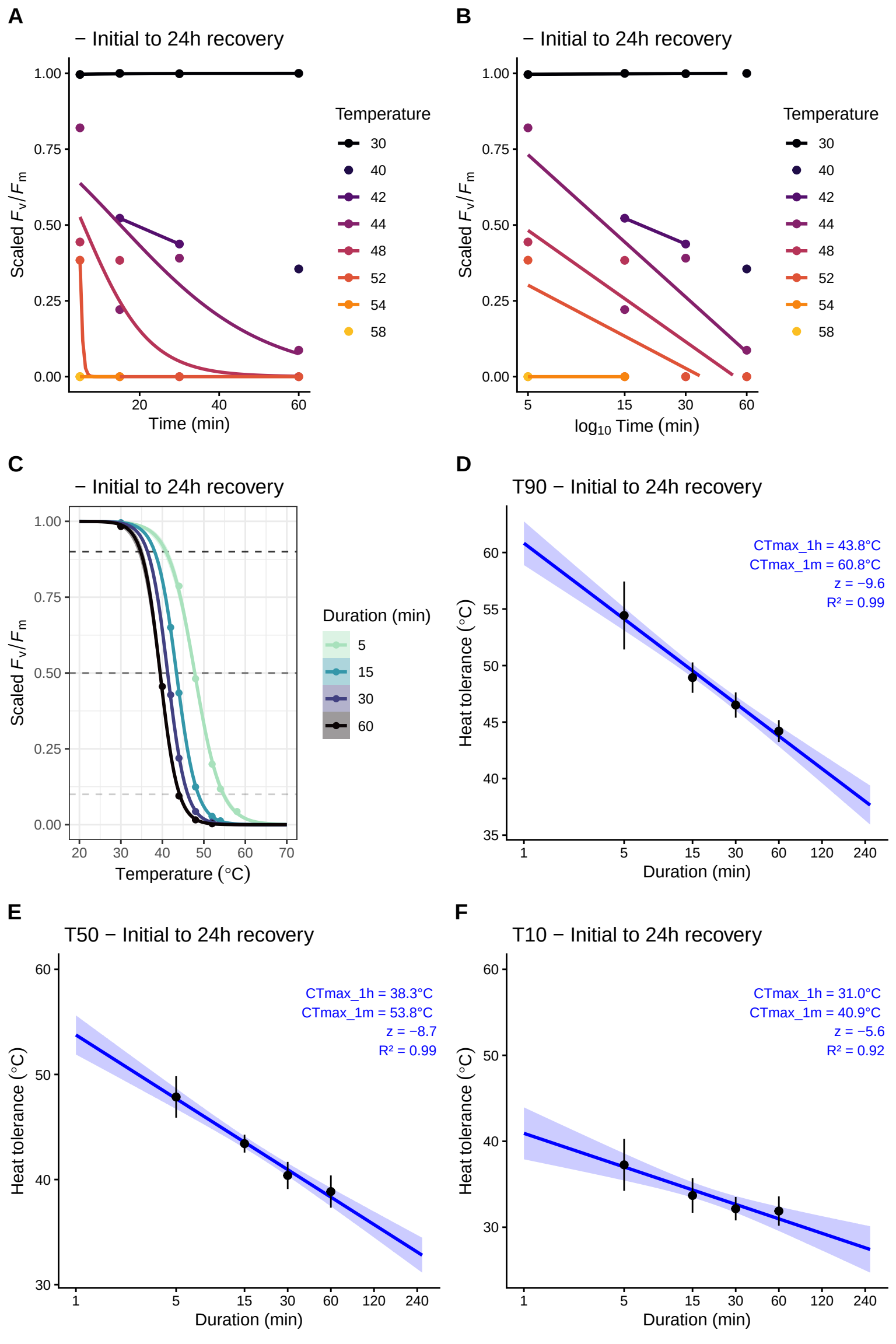

A
− Initial to 2h post−heat
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Initial to 2h post−heat
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Initial to 2h post−heat
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T90 − Initial to 2h post−heat
CTmax_1h = 44.4°C
CTmax_1m = 65.4°C
z = −11.8
R² = 0.94
Heat tolerance (°C)
Duration (min)
E
T50 − Initial to 2h post−heat
CTmax_1h = 38.8°C
CTmax_1m = 50.5°C
z = −6.6
R² = 0.94
Heat tolerance (°C)
Duration (min)
F
T10 − Initial to 2h post−heat
CTmax_1h = 31.5°C
CTmax_1m = 34.1°C
z = −1.4
R² = 0.34
Heat tolerance (°C)
Duration (min)

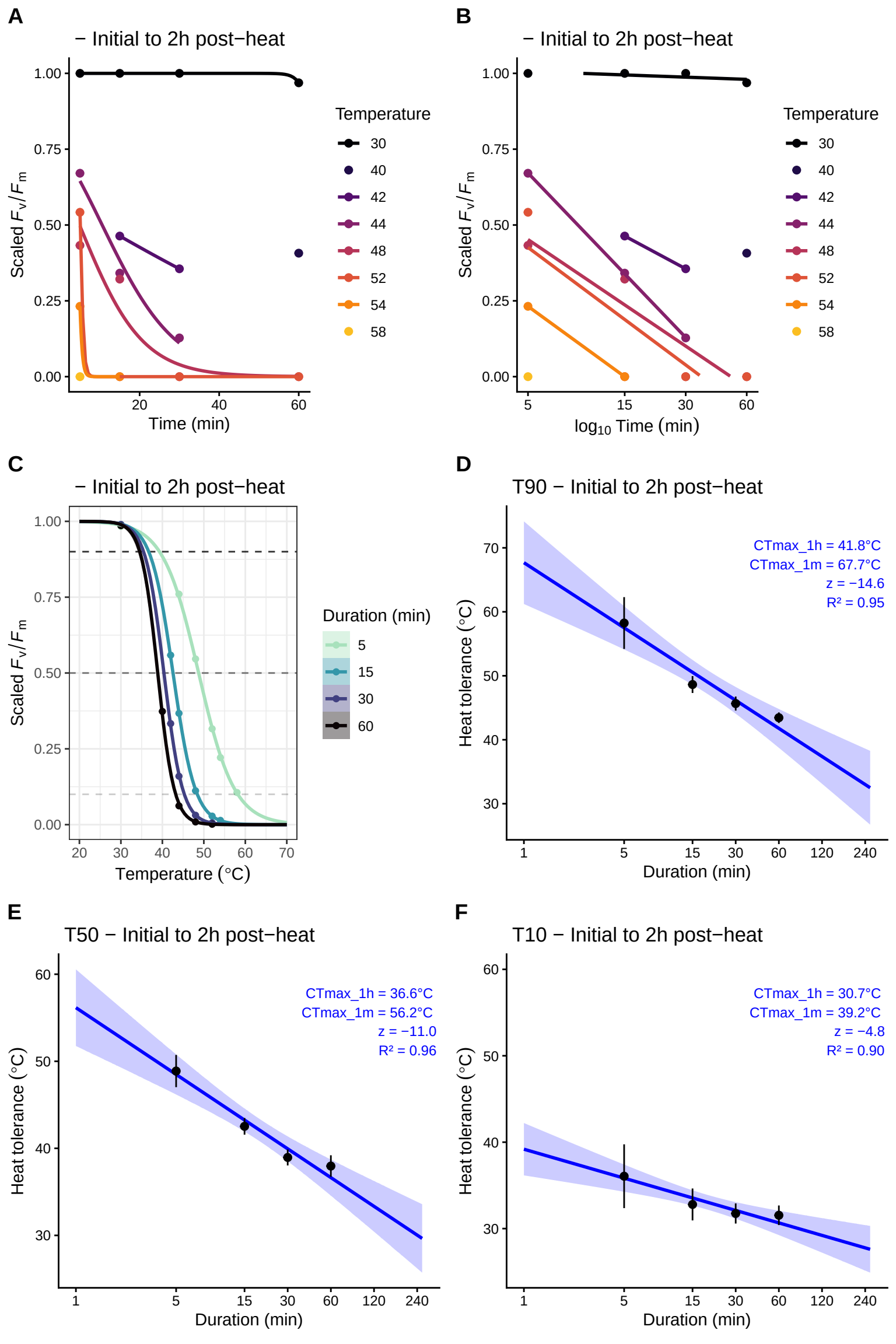

A
− Absolute Fv/Fm values
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Absolute Fv/Fm values
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Absolute Fv/Fm values
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T0.3 − Absolute Fv/Fm values
CTmax_1h = 38.6°C
CTmax_1m = 53.3°C
z = −8.3
R² = 0.97
Heat tolerance (°C)
Duration (min)

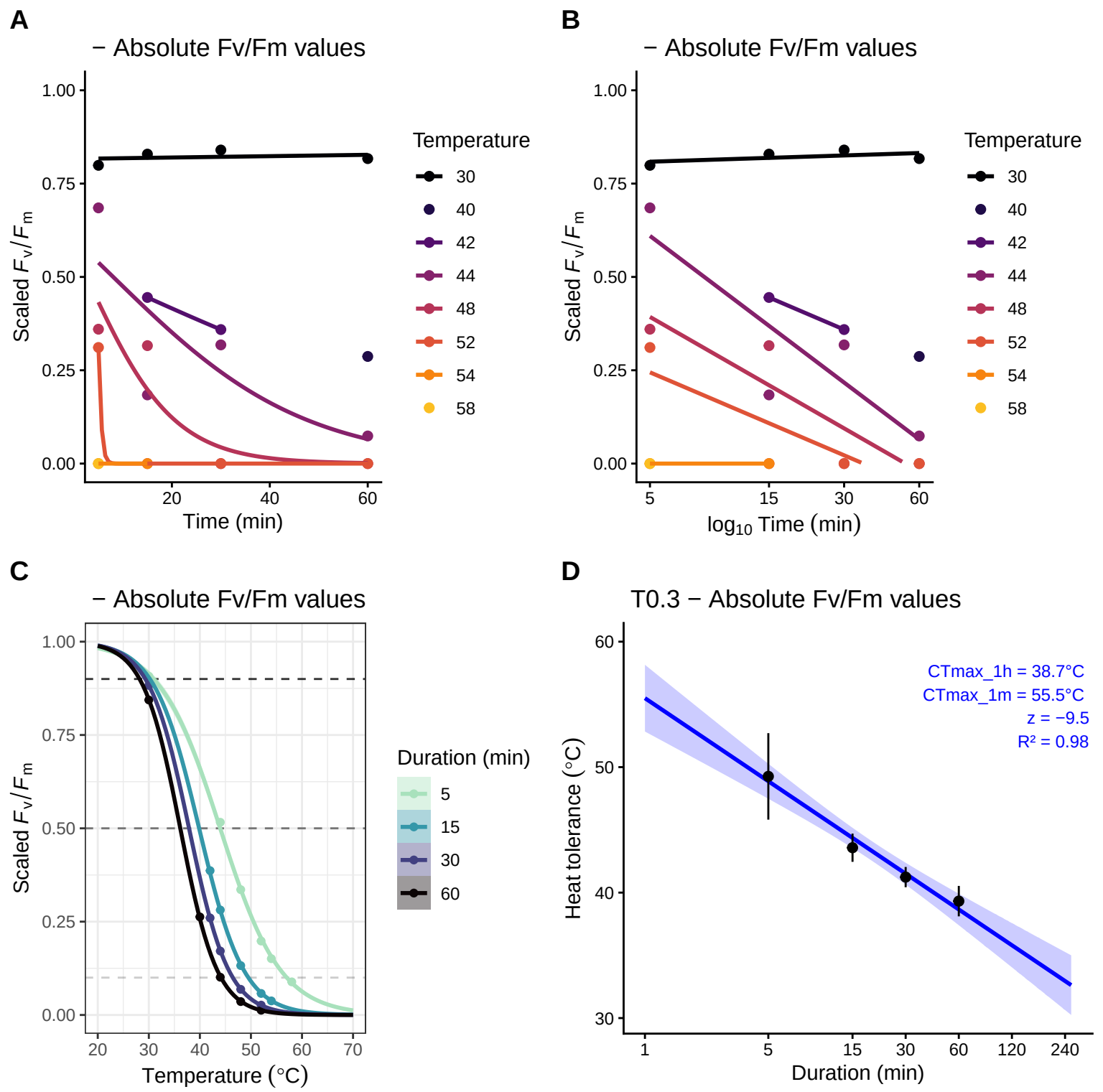

A
− Initial to 24h recovery
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Initial to 24h recovery
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Initial to 24h recovery
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T90 − Initial to 24h recovery
CTmax_1h = 43.2°C
CTmax_1m = 53.6°C
z = −5.8
R² = 0.99
Heat tolerance (°C)
Duration (min)
E
T50 − Initial to 24h recovery
CTmax_1h = 38.3°C
CTmax_1m = 49.7°C
z = −6.4
R² = 0.96
Heat tolerance (°C)
Duration (min)
F
T10 − Initial to 24h recovery
CTmax_1h = 32.2°C
CTmax_1m = 37.3°C
z = −2.9
R² = 0.97
Heat tolerance (°C)
Duration (min)


A
− Initial to 2h post−heat
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Initial to 2h post−heat
Scaled F_v/F_m
$\log_{10}$ Time (min)
C
− Initial to 2h post−heat
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T90 − Initial to 2h post−heat
CTmax_1h = 40.7°C
CTmax_1m = 65.5°C
z = −14.0
R² = 0.95
Heat tolerance (°C)
Duration (min)
E
T50 − Initial to 2h post−heat
CTmax_1h = 35.0°C
CTmax_1m = 54.7°C
z = −11.1
R² = 0.95
Heat tolerance (°C)
Duration (min)
F
T10 − Initial to 2h post−heat
CTmax_1h = 30.3°C
CTmax_1m = 37.4°C
z = −4.0
R² = 0.90
Heat tolerance (°C)
Duration (min)


A
− Absolute Fv/Fm values
Scaled F_v/F_m
1.00
0.75
0.50
0.25
0.00
20
40
60
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Absolute Fv/Fm values
Scaled F_v/F_m
1.00
0.75
0.50
0.25
0.00
5
15
30
60
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Absolute Fv/Fm values
Scaled F_v/F_m
1.00
0.75
0.50
0.25
0.00
20
30
40
50
60
70
Temperature (°C)
Duration (min)
5
15
30
60
D
T0.3 − Absolute Fv/Fm values
CTmax_1h = 38.3°C
CTmax_1m = 49.6°C
z = −6.4
R² = 0.99
Heat tolerance (°C)
60
50
40
1
5
15
30
60
120
240
Duration (min)


A
− Initial to 24h recovery
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Initial to 24h recovery
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Initial to 24h recovery
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T90 − Initial to 24h recovery
CTmax_1h = 43.8°C
CTmax_1m = 60.8°C
z = −9.6
R² = 0.99
Heat tolerance (°C)
Duration (min)
E
T50 − Initial to 24h recovery
CTmax_1h = 38.3°C
CTmax_1m = 53.8°C
z = −8.7
R² = 0.99
Heat tolerance (°C)
Duration (min)
F
T10 − Initial to 24h recovery
CTmax_1h = 31.0°C
CTmax_1m = 40.9°C
z = −5.6
R² = 0.92
Heat tolerance (°C)
Duration (min)


A
− Initial to 2h post−heat
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Initial to 2h post−heat
Scaled F_v/F_m
$\log_{10}$ Time (min)
C
− Initial to 2h post−heat
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T90 − Initial to 2h post−heat
CTmax_1h = 41.8°C
CTmax_1m = 67.7°C
z = −14.6
R² = 0.95
Heat tolerance (°C)
Duration (min)
E
T50 − Initial to 2h post−heat
CTmax_1h = 36.6°C
CTmax_1m = 56.2°C
z = −11.0
R² = 0.96
Heat tolerance (°C)
Duration (min)
F
T10 − Initial to 2h post−heat
CTmax_1h = 30.7°C
CTmax_1m = 39.2°C
z = −4.8
R² = 0.90
Heat tolerance (°C)
Duration (min)


A
− Absolute Fv/Fm values
Scaled F_v/F_m
Time (min)
Temperature
30
40
42
44
48
52
54
58
B
− Absolute Fv/Fm values
Scaled F_v/F_m
$\log_{10}$ Time (min)
Temperature
30
40
42
44
48
52
54
58
C
− Absolute Fv/Fm values
Scaled F_v/F_m
Temperature (°C)
Duration (min)
5
15
30
60
D
T0.3 − Absolute Fv/Fm values
CTmax_1h = 38.7°C
CTmax_1m = 55.5°C
z = −9.5
R² = 0.98
Heat tolerance (°C)
Duration (min)